# Athena Sy Kuzuhara

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## EDUCATION

- Bachelor of Science in Computer Science, **University of Southern California,** August 2023-December 2026
    - Relevant Coursework: Introduction to Programming, Introduction to Embedded Systems, Data Structures and Object Oriented Design, Principles of Software Development, Introduction to Algorithms and Theory of Computing, Introduction to Computer Systems, Introduction to Artificial Intelligence, Introduction to Internetworking, Introduction to Operating Systems, Software Engineering, Introduction to Machine Learning
- Master of Science in Computer Science,**University of Southern California,** August 2025-December 2026
    - Relevant Coursework: Database Systems, Analysis of Algorithms, Parallel and Distributed Programming, 3-D Graphics and Rendering, Machine Learning, Deep Learning, Scientific Computing and Visualization

## SKILLS

C/C++ | Python | Java | Javascript | Typescript | Go | Git | Linux | Data Structures | Machine Learning | Grafana | Bash

## EXPERIENCE

**Cerebras Systems, Sunnyvale, CA | Software Engineering Intern | May 2025-August 2025**

- Set up and configured Grafana dashboards to monitor and visualize key system metrics including disk utilization and CPU usage across company servers, improving performance visibility and alert response times
- Developed an opcode calculator tool that processes opcode and source inputs to generate destination values, enabling senior engineers to verify simulator accuracy and validate instruction execution logic

**Cerebras Systems, Sunnyvale, CA | Software Engineering Intern | May 2024-August 2024**

- Automated legacy code generation using Python, reducing the codebase by hundreds of lines and significantly improving maintainability and development efficiency
- Collaborated with senior engineers to identify and resolve critical bugs while optimizing testing workflows for faster, more reliable verification

**Laboratory of Dr. Stephen J. Read, Los Angeles, CA | Research Assistant, | October 2023-December 2024**

- Used Emergent neural network software to model depression and decision making
- Read and analyzed research on computational cognitive neuroscience
- Trained and tested neural network model of depression changing parameters that modeled chemicals in brain
- Used generative AI to develop research techniques
- Assisted in optimizing software in Go programming language to meet research needs

## PROJECTS

**Team Hearing**

- Built a full-stack accessibility platform (React + Spring Boot) to enhance music perception for users with hearing impairments
- Implemented and integrated audio/hearing modules, supporting scalable addition of new signal-processing features, designing a modular architecture to reduce code dependencies and enable rapid feature expansion

**Parallelizing a Text to Speech Pipeline**

- Designed and implemented a parallelized neural text-to-speech pipeline using PyTorch to reduce inference latency
- Applied pipeline parallelism to overlap preprocessing, acoustic generation, and waveform synthesis stages and developed chunk-based parallel synthesis, enabling concurrent processing of text segments across multiple workers

**Predicting COVID-19 Loss and Recovery**

- Modeled district-level learning loss and recovery using large-scale education datasets
- Implemented Ridge/Lasso regression to predict test score changes and quantify pandemic impact on academic performance
- Built classification models (Logistic Regression, SVM) to predict recovery outcomes across school districts